Forensic Baseline Report

Drone Forensic Decision and Orchestration Framework

Case Identifier:	ft-df003-2017-ios
Operator:	Floris
Analysis Started:	2026-06-18 10:12:52
Analysis Completed:	2026-06-18 10:17:29
Evidence Directory:	C:\Users\Floris\Documents\dji_drones\train_phantom_3\df003\2017\ios
Output Directory:	C:\Users\Floris\Documents\dji_drones\train_phantom_3\df003\2017\dfdof_output\ft-df003-2017-ios
Evidence Sources:	3
Pipeline Version:	DFDOF v1.0 (proof of concept)

This report is an automated forensic baseline produced by DFDOF. It is not a court-ready expert report. All findings are bounded by the data available and must be interpreted by a qualified forensic examiner.The populated case output directory is the authorative source and provides the full overview of the pipeline's findings.

1. Evidence Intake and Chain of Custody

The following input evidence files were ingested. SHA-256 and SHA-1 hashes were computed at intake to establish integrity baselines. No modification to original evidence files is performed by DFDOF.

File	Classification	Acquisition	Size	Hash Timestamp
df003_external.001	drone_sd	physical	7.4 GB	2026-06-18 10:13:23
df003_flight.zip	controller_ios	logical	207.5 MB	2026-06-18 10:13:24
df003_internal.001	drone_flight_storage	physical	3.7 GB	2026-06-18 10:13:36

Evidence Source Path

C:\Users\Floris\Documents\dji_drones\train_phantom_3\df003\2017\ios

Cryptographic Hashes

df003_external.001	drone_sd	physical
SHA-256:	`b88f6f60a1ad278ca0a0c246c2c09f169b4c94fd281d90dbefbb867206f83acc`	
SHA-1:	`6b4905eb6cba14b8dcc1ea78b07a7f176b1b8115`	

df003_flight.zip	controller_ios	logical
SHA-256:	`bccdb19faad0df85422bcb4f7efd695ca970fc4c0db5841a576898108b8f4e35`	
SHA-1:	`a09cca4b196f4b5d471d073c14d4829291766417`	

df003_internal.001	drone_flight_storage	physical
SHA-256:	`14c40dc6243987d7bc631e71ec8f1581717e408dfb21dc88aaa437962e77216c`	
SHA-1:	`f321d1a6d334c96ee6fbb446f4877a9e0664ef33`	

2. Pipeline Execution Summary

Completed Phases

1. p1_provenance
2. p2_image_parsing
3. p3_artefact_extraction
4. p4_decision_and_orchestration
5. p5_normalisation_and_anomaly_checking
6. p6_multisource_correlation
7. p7_analysis_and_validation
8. p8_automated_reporting

Anomaly Flags

1. [p2 - controller android]: source evidence not found
2. [p3 - controller android]: source evidence not found
3. [p3 - controller ios - drone logs]: no artefacts found
4. [p3 - drone flight storage - drone logs]: empty file moved to _rejected: FLY007.DAT
5. [p4 - controller android]: source evidence not found

Tool Execution Status

Tool	Invocations	Successes	Status
datcon	7	7	[OK]
exiftool	4	3	[~] Partial
fls	2	2	[OK]
icat	10	10	[OK]
mmls	2	2	[OK]
parser_ios	1	1	[OK]
txtlogtocsvtool	2	2	[OK]

Processing Coverage Score

Metric	Result
Evidence Sources Detected	3 / 4
Artefact Categories With Data	7 / 7
Flights With Primary Correlation	1 / 8
Tools Succeeded	6 / 7

3. Device and Account Identification

Field	Value	Confidence	Sources
Installed DJI App(s)	com.dji.go com.dji.pilot	[K] Controller-only	1
Controller Device	iPad mini 4	[K] Controller-only	1
Drone Serial Number	0000000000000000	[K] Controller-only	2
Drone Model	P3 Standard P3I1 P3S	[!] Inconsistent	9
Controller Device	iPad5,1__weibosdk__003143000__iphone__os 10.3.1 iPad5,1__weibosdk__003143000__iphone__os 10.3.2	[!] Inconsistent	2
DJI Application Version	3.1.12 4.1.3	[!] Inconsistent	4
Account E-Mail	droneforensics@vtoinc.com	[K] Controller-only	2
Product Type	2 7	[!] Inconsistent	2
Last Country Code	US	[K] Controller-only	1
Last App Launch	2017-06-28T19:40:33Z 2017-06-28T20:20:50Z	[!] Inconsistent	2
Last Flight Date	1970-01-01T00:00:00Z	[K] Controller-only	2

4. Flight Analysis - Flight 01

Correlation and Flight Window

Status:	Unmatched (no drone log corroboration)
Drone Log SHA-256:	`0f033f03fa12fd04fa7036c7a361803088372e81a1d2a07663a9b2616695d281`
Drone Log Name:	FLY000.csv
Start:	2017-06-28T17:00:52.860000+00:00
End:	2017-06-28T17:11:58.100000+00:00
Duration:	665.2 s

Flight Metrics

Peak Height:	-
Distance Travelled:	-
Video Recording:	No
Photo Capture:	No

Fly Modes Observed (Chronological)

1. 0

Key Events

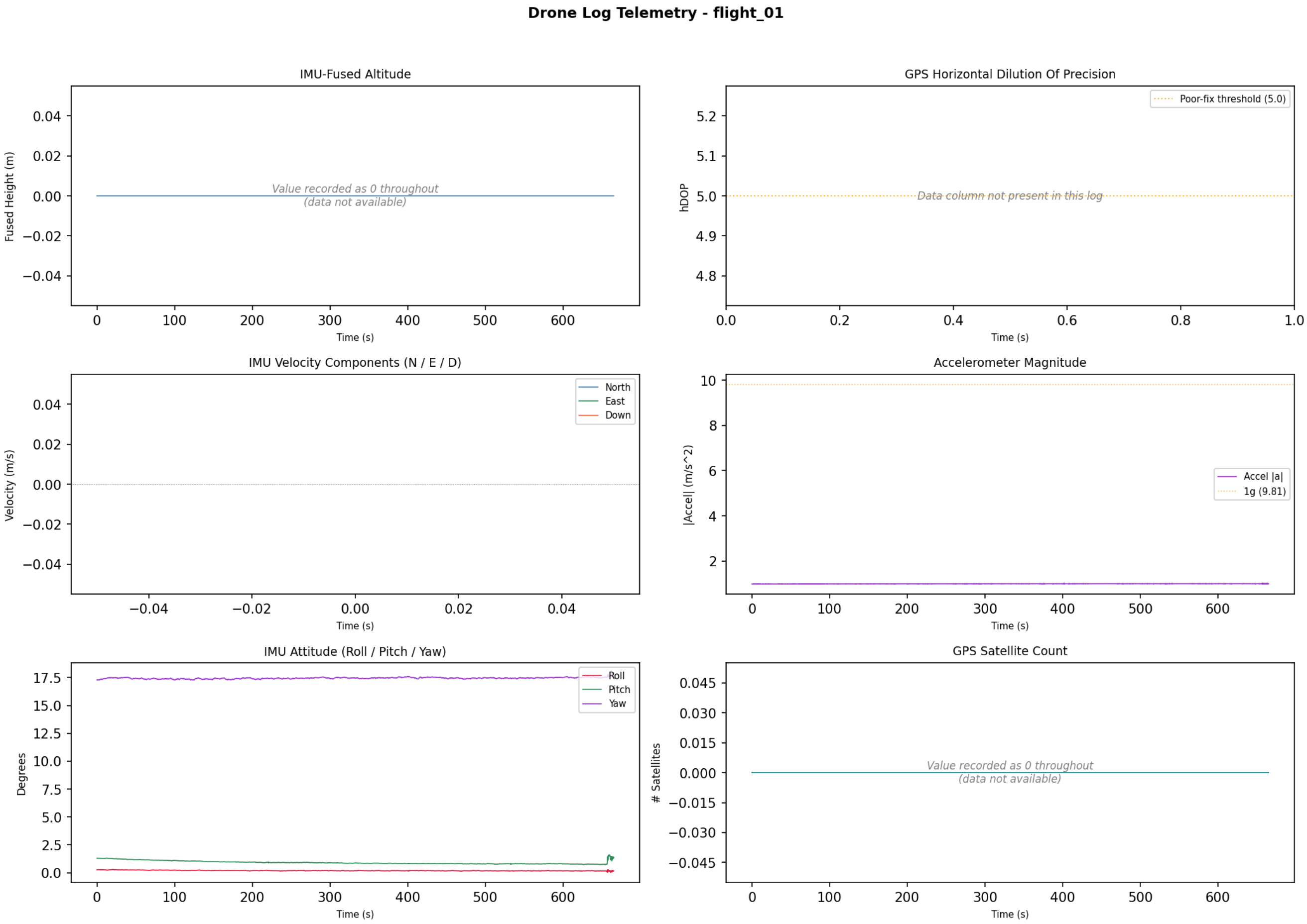
Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Reached peak height	-0.0 m
2017-06-28 17:00:53	drone_flight_storage:dl	Log started	P3I1
2017-06-28 17:11:55	drone_flight_storage:dl	Log ended	

Correlated Artefacts

Possibly correlated (indirect indicators):

drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:e034453ea8d0e3db7eaf530cced43bcbc7036483b9e01ccb19fbfd2d6287c6c7	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 02

Correlation and Flight Window

Status:	Unmatched (no drone log corroboration)
Drone Log SHA-256:	`b2cc4417cd0f099e3a94bfec48ae02cfc77bdf1bd73e5c9710554ef4ec81ab38`
Drone Log Name:	FLY001.csv
Start:	2017-06-28T17:14:36.693000+00:00
End:	2017-06-28T17:24:56.415000+00:00
Duration:	619.7 s

Flight Metrics

Peak Height:	-
Distance Travelled:	-
Video Recording:	No
Photo Capture:	No

Fly Modes Observed (Chronological)

1. 0

Key Events

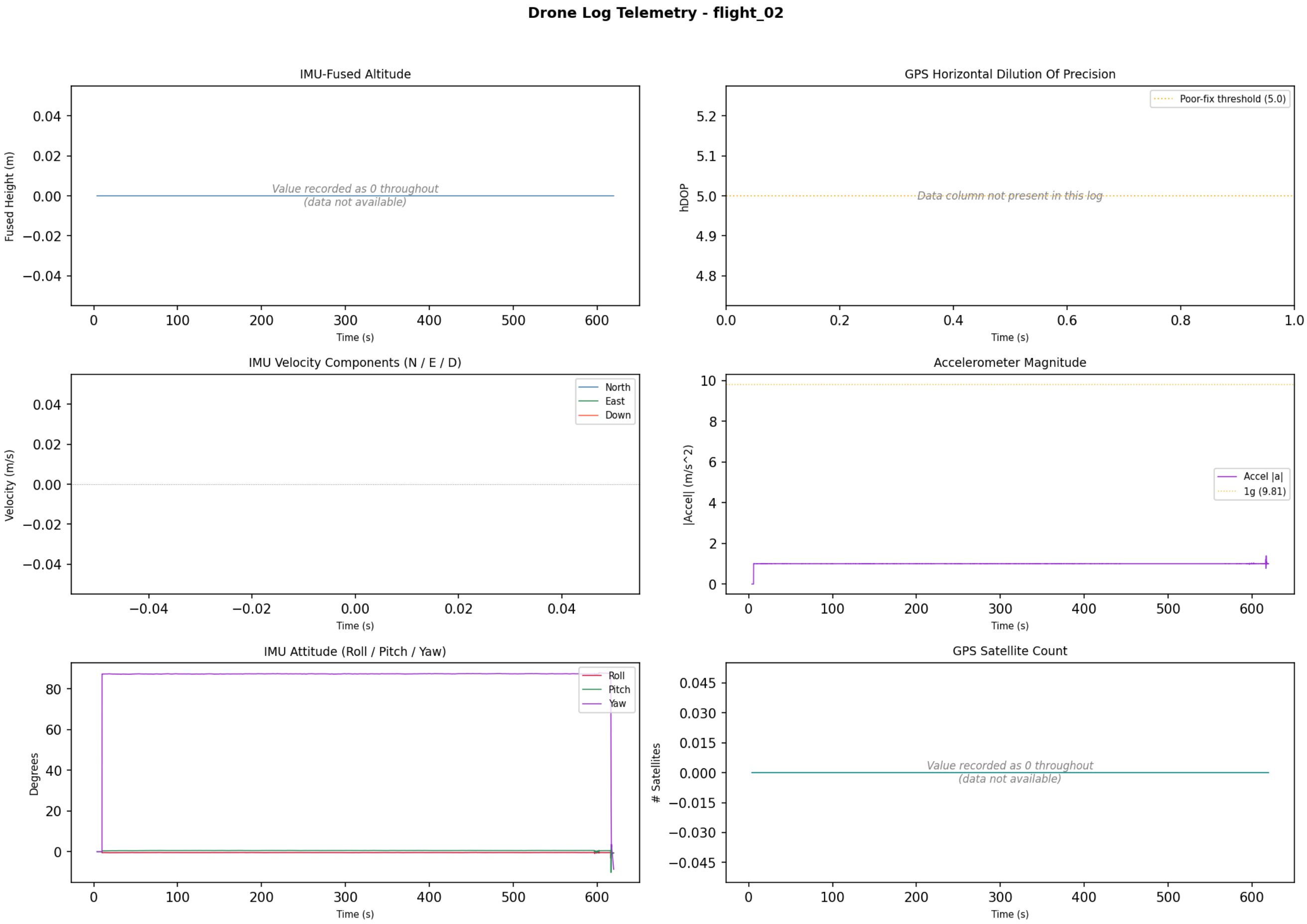
Timestamp (UTC)	Source	Event	Details
2017-06-28 17:15:14	drone_flight_storage:dl	Log started	P3S
2017-06-28 17:24:56	drone_flight_storage:dl	Log ended	
3236-01-12 23:59:47	drone_flight_storage:dl	Reached peak height	-0.0 m

Correlated Artefacts

Possibly correlated (indirect indicators):

drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:e034453ea8d0e3db7eaf530cced43bcbc7036483b9e01ccb19fbfd2d6287c6c7	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 03

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: `373f06f7c3a551a0626de74b0978df4d035aca01e9651de3fd3fa0a78bd5acb7`
Drone Log Name: FLY002.csv
Start: 2017-06-28T18:12:43.487000+00:00
End: 2017-06-28T18:21:54.690000+00:00
Duration: 551.2 s

Flight Metrics

Peak Height: -
Distance Travelled: -
Video Recording: No
Photo Capture: No

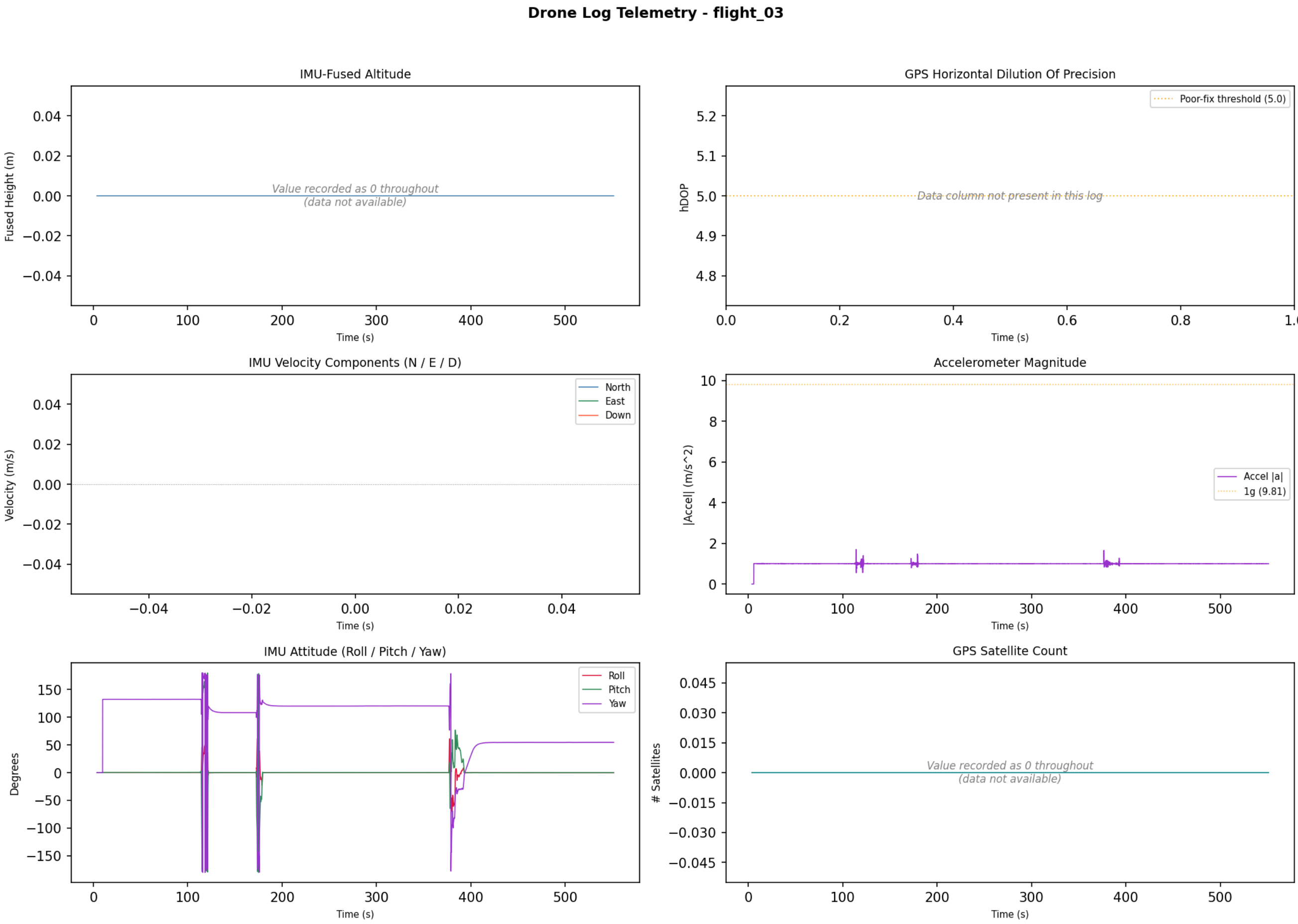
Fly Modes Observed (Chronological)

1. 0

Key Events

Timestamp (UTC)	Source	Event	Details
2017-06-28 18:20:47	drone_flight_storage:dl	Log started	P3S
2017-06-28 18:21:51	drone_flight_storage:dl	Log ended	
3236-01-12 23:59:47	drone_flight_storage:dl	Reached peak height	-0.0 m

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 04

Correlation and Flight Window

Status:	Unmatched (no drone log corroboration)
Drone Log SHA-256:	`ebaffbc6750e8b1935676eb51e28ef62ba1f6e7d74ebeadf98125bf2f2ec2a1f`
Drone Log Name:	FLY003.csv
Start:	2017-06-28T18:27:38.290000+00:00
End:	2017-06-28T18:34:18.570000+00:00
Duration:	400.3 s

Flight Metrics

Peak Height:	-
Distance Travelled:	-
Video Recording:	No
Photo Capture:	No

Fly Modes Observed (Chronological)

1. 0

Key Events

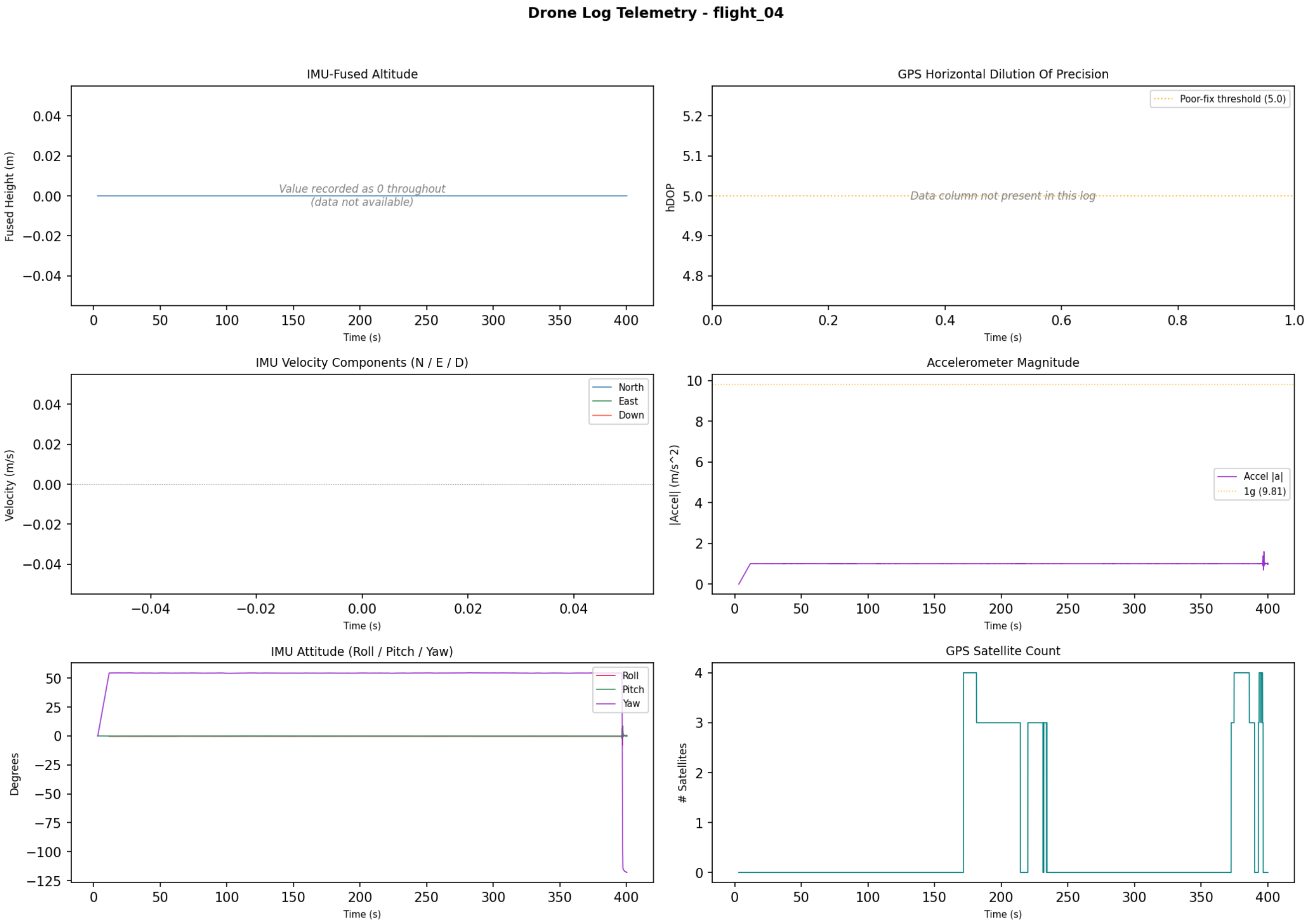
Timestamp (UTC)	Source	Event	Details
2017-06-28 18:27:39	drone_flight_storage:dl	Log started	P3S
2017-06-28 18:27:39	drone_flight_storage:dl	Reached peak height	-0.0 m
2017-06-28 18:34:18	drone_flight_storage:dl	Log ended	

Correlated Artefacts

Possibly correlated (indirect indicators):

drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:e034453ea8d0e3db7eaf530cced43bcbc7036483b9e01ccb19fbfd2d6287c6c7	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 05

Correlation and Flight Window

Status:	Matched (primary GPS+temporal)
Confidence:	high
GPS Overlap:	104.5 s
Median GPS Distance:	0.24 m
Overlap:	100%
GPS Match Rate:	100%
Flight Record SHA-256:	016e5b03b09fffd4158b51d3317e192496b8e5e5f5738996db15c0938a8f049f
Flight Record Name:	DJIFlightRecord_2017-06-29__10-30-47_.csv
Start:	2017-06-29T16:30:47.317000+00:00
End:	2017-06-29T16:32:31.834000+00:00
Duration:	104.5 s
Drone Log SHA-256:	21fab79500aa2a2d097ba682e2f943de74cc03911a269c60f9fd3196ee0fa126
Drone Log Name:	FLY004.csv
Start:	2017-06-29T16:25:23.810000+00:00
End:	2017-06-29T16:42:37.705000+00:00
Duration:	1033.9 s

Flight Metrics

Peak Height:	5.2 m
Distance Travelled:	16.49 m
Video Recording:	No
Photo Capture:	No
Photos Taken:	0
Recording Duration:	0.0 s
Record Complete:	Yes

Fly Modes Observed (Chronological)

1. 2
2. GPS_Atti
3. AutoTakeoff
4. 1
5. 3
6. 0

Key Events

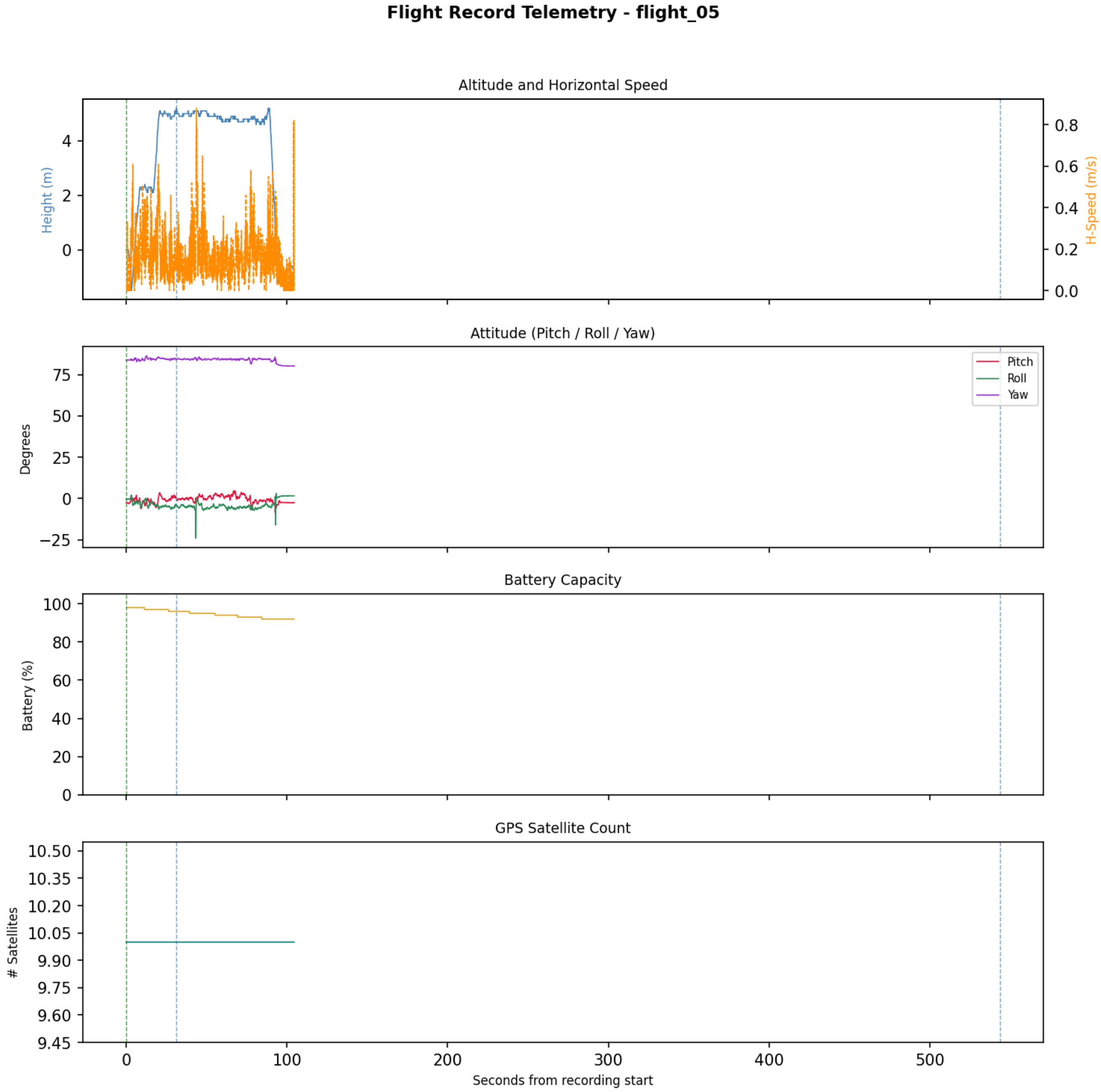
Timestamp (UTC)	Source	Event	Details
2017-06-29 16:25:47	drone_flight_storage:dl	Log started	P3S
2017-06-29 16:30:47	ctrl_ios:fr (v3.1.12)	Log started	SN: 0000000000000000; App: 3.1.12; P3 Standard
2017-06-29 16:30:47	ctrl_ios:fr (v3.1.12)	Motor turned on	on
2017-06-29 16:30:48	ctrl_ios:fr (v3.1.12)	Record mode changed	No
2017-06-29 16:31:18	ctrl_ios:fr (v3.1.12)	Reached peak height	5.2 m
2017-06-29 16:32:31	ctrl_ios:fr (v3.1.12)	Log ended	Dist home: 0.3 m; Height: -0.7 m; Rec: 0.0 s
2017-06-29 16:39:51	drone_flight_storage:dl	Reached peak height	45.6 m
2017-06-29 16:42:34	drone_flight_storage:dl	Log ended	

Correlated Artefacts

Possibly correlated (indirect indicators):

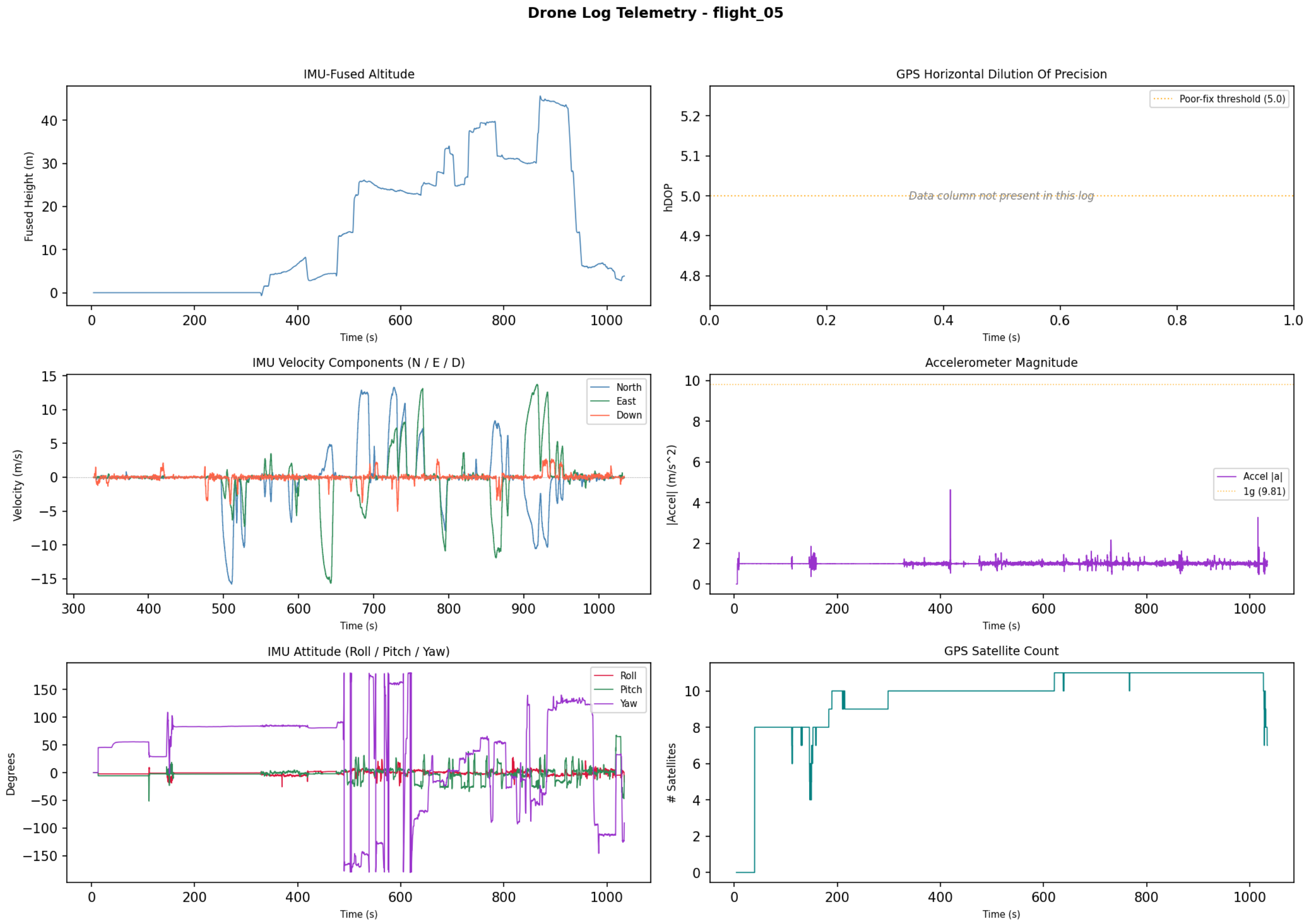
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:e034453ea8d0e3db7eaf530cced43bcbc7036483b9e01ccb19fbfd2d6287c6c7	

Flight Record - Sensor Telemetry



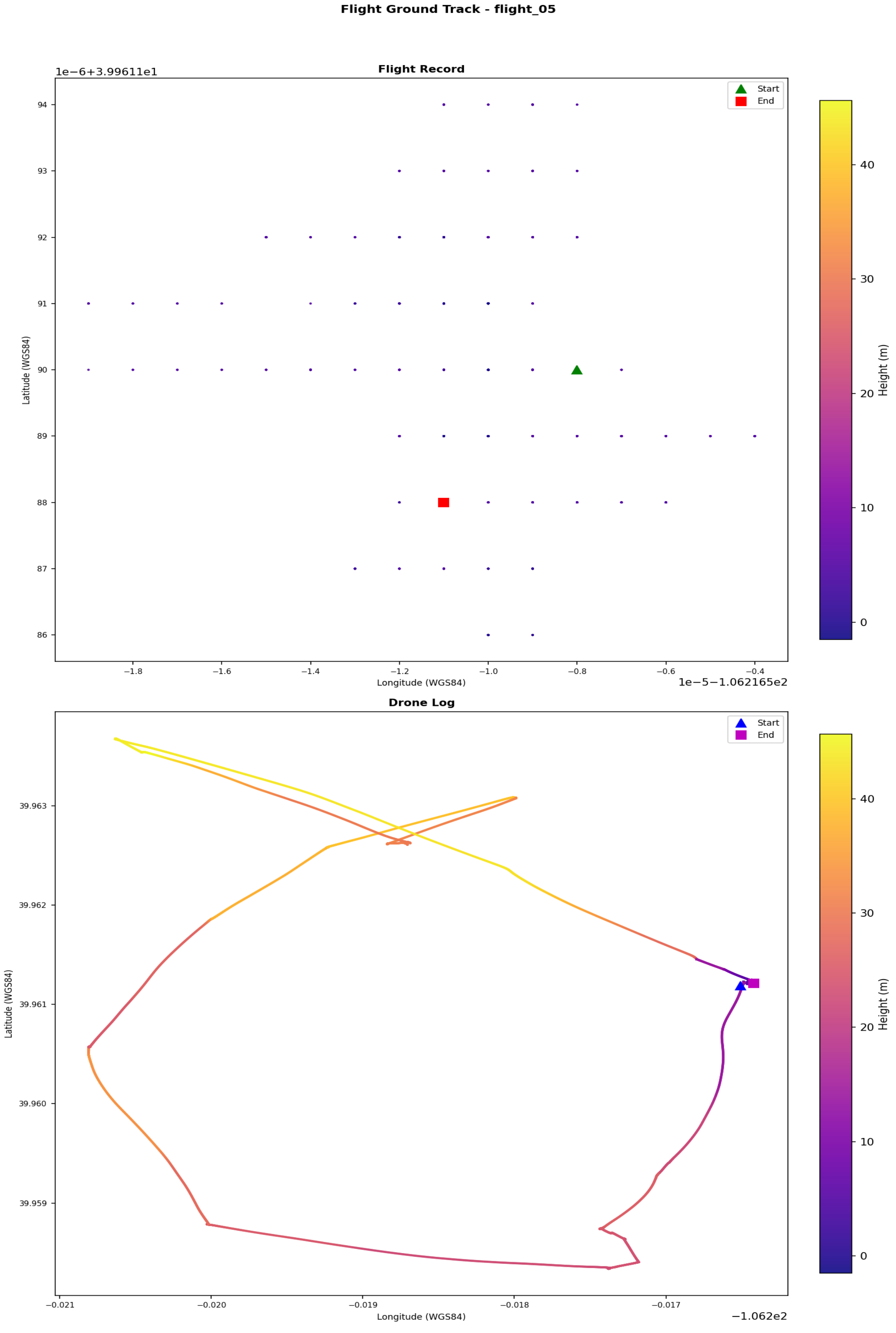
GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY004.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY004.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 06

Correlation and Flight Window

Status:	Unmatched (no drone log corroboration)
Flight Record SHA-256:	2547d057517715c42f54a1348018002c9ecdd94acbedddcdbaf469f7e9bfc93c
Flight Record Name:	DJIFlightRecord_2017-06-29__10-32-33_.csv
Start:	2017-06-29T16:32:33.918000+00:00
End:	2017-06-29T16:42:25.611000+00:00
Duration:	591.7 s

Flight Metrics

Peak Height:	43.0 m
Distance Travelled:	2011.24 m
Video Recording:	Yes
Photo Capture:	No
Photos Taken:	0
Recording Duration:	544.0 s
Record Complete:	Yes

Fly Modes Observed (Chronological)

1. GPS_Atti
2. AssistedTakeoff

Key Events

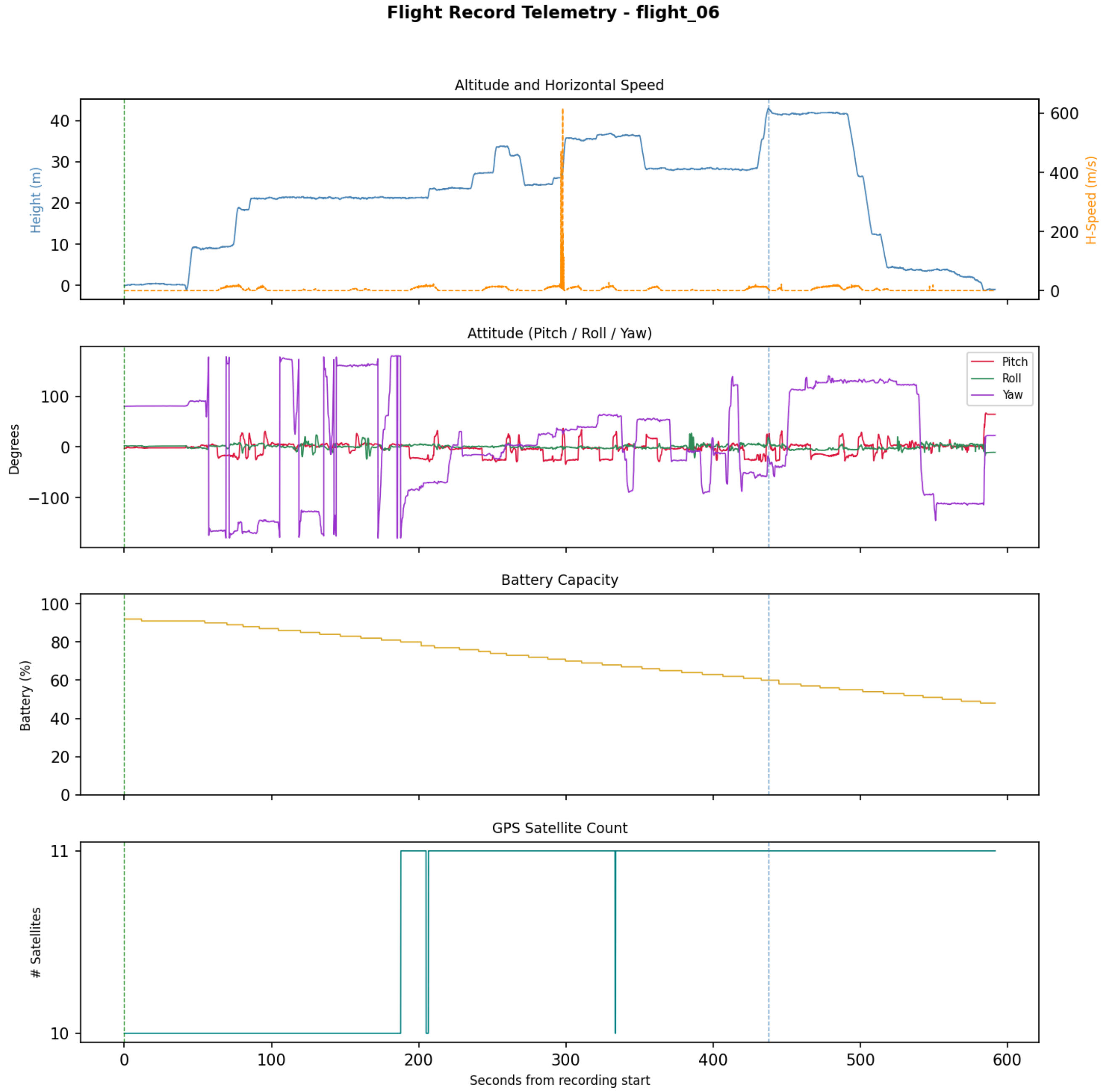
Timestamp (UTC)	Source	Event	Details
2017-06-29 16:32:33	ctrl_ios:fr (v3.1.12)	Log started	SN: 0000000000000000; App: 3.1.12; P3 Standard
2017-06-29 16:32:33	ctrl_ios:fr (v3.1.12)	Motor turned on	on
2017-06-29 16:32:34	ctrl_ios:fr (v3.1.12)	Record mode changed	No
2017-06-29 16:33:22	ctrl_ios:fr (v3.1.12)	Record mode changed	Starting
2017-06-29 16:39:51	ctrl_ios:fr (v3.1.12)	Reached peak height	43.0 m
2017-06-29 16:42:25	ctrl_ios:fr (v3.1.12)	Log ended	Dist home: 0.6 m; Height: -1.0 m; Rec: 544.0 s

Correlated Artefacts

Possibly correlated (indirect indicators):

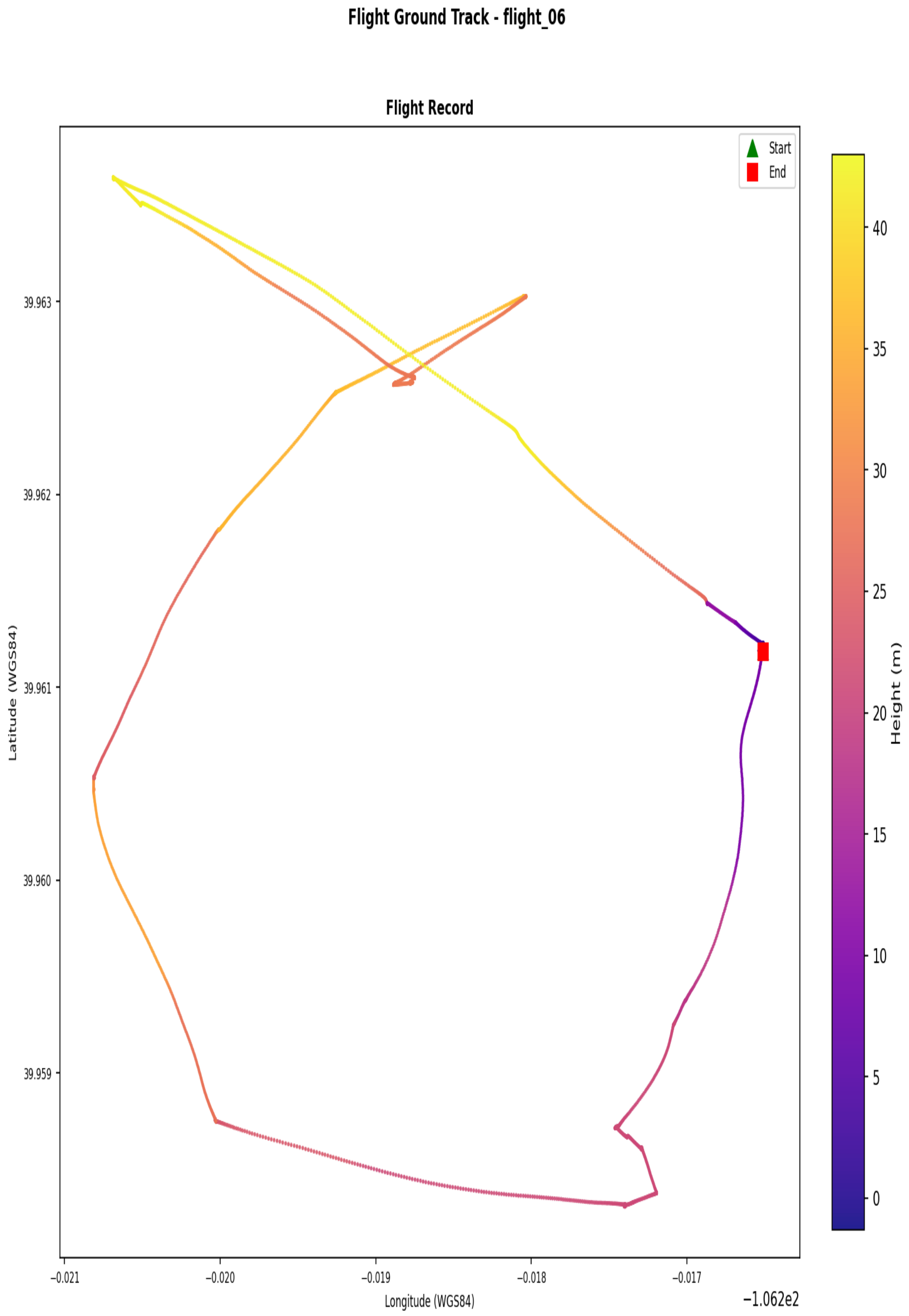
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:e034453ea8d0e3db7eaf530cced43bcbc7036483b9e01ccb19fbfd2d6287c6c7	

Flight Record - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m).

Coordinates are WGS84. No base map. Colour represents relative height (metres).

4. Flight Analysis - Flight 07

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: `bcdeb039bb63d782cf8ad16922f267fd4d18b79648dfe468b9901ca550330a06`
Drone Log Name: FLY005.csv
Start: 2017-06-29T19:51:40.032000+00:00
End: 2017-06-29T20:18:35.530000+00:00
Duration: 1615.5 s

Flight Metrics

Peak Height: 40.89038704705247 m
Distance Travelled: 2727.6109987801456 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

1. 2
2. 1
3. 3
4. 0

Key Events

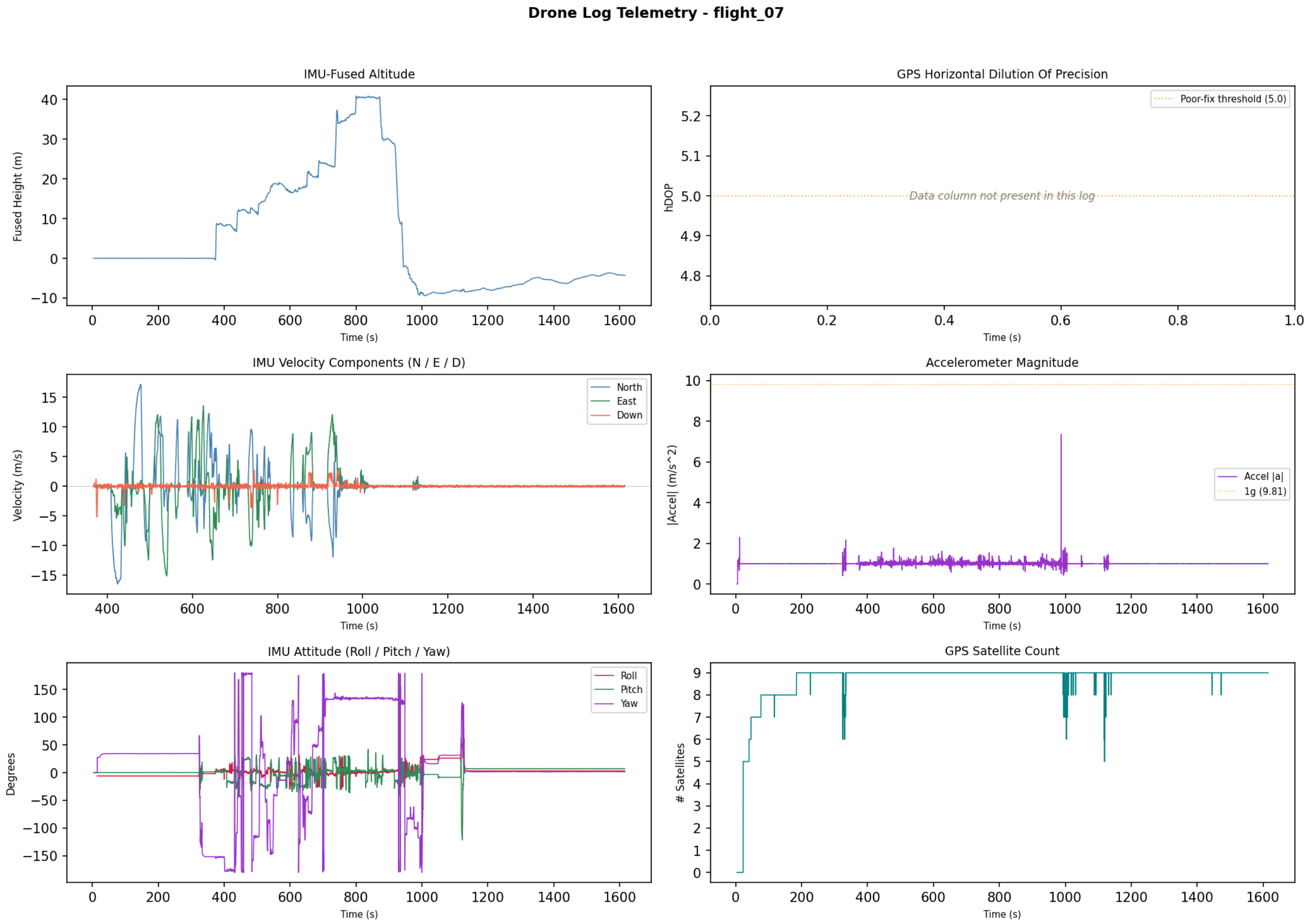
Timestamp (UTC)	Source	Event	Details
2017-06-29 19:51:52	drone_flight_storage:dl	Log started	P3S
2017-06-29 20:04:58	drone_flight_storage:dl	Reached peak height	40.9 m
2017-06-29 20:18:32	drone_flight_storage:dl	Log ended	

Correlated Artefacts

Possibly correlated (indirect indicators):

drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:e034453ea8d0e3db7eaf530cced43bcbc7036483b9e01ccb19fbfd2d6287c6c7	

Drone Log - Sensor Telemetry



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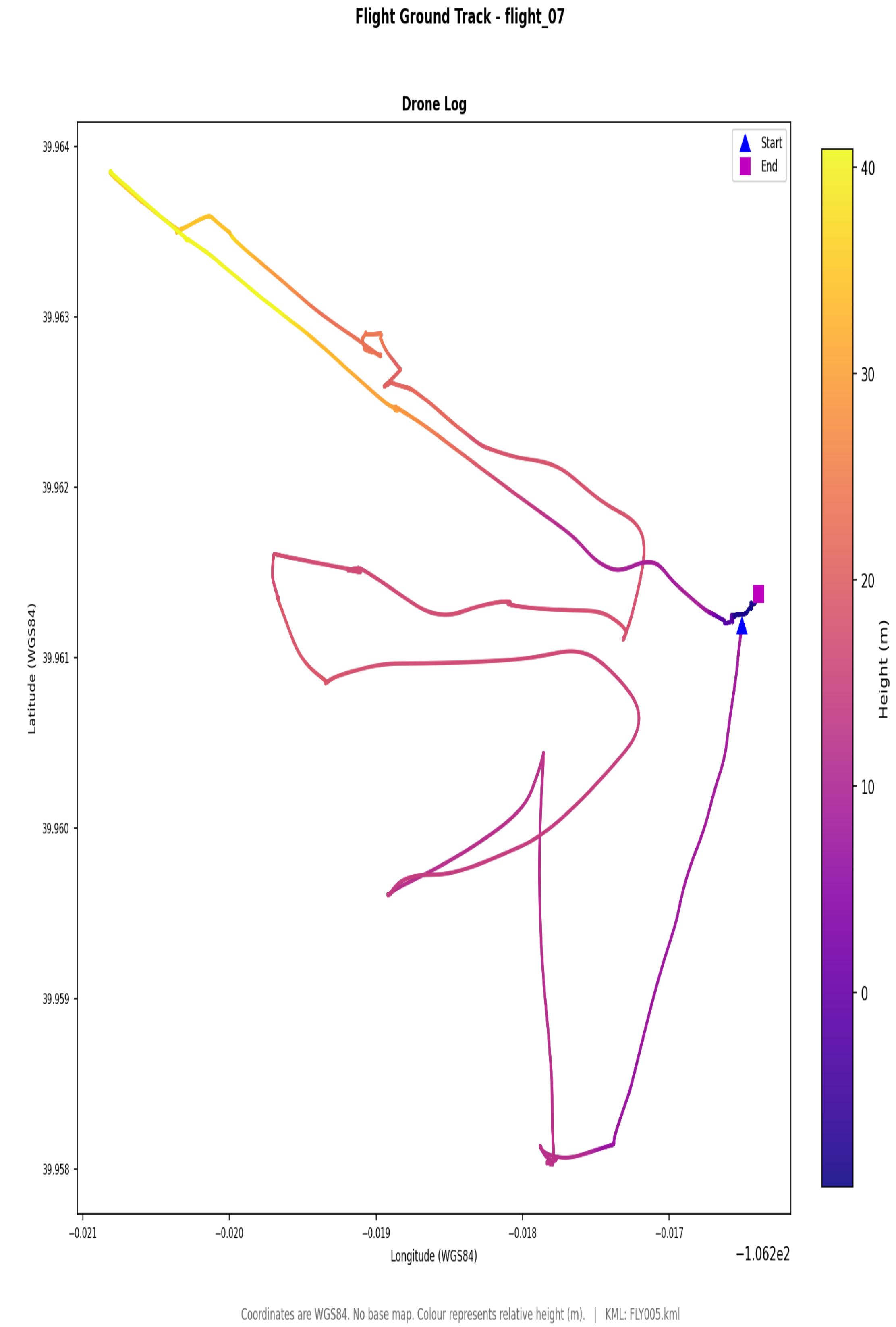
Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY005.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 08

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: `24d395909e47a8d9bd09e8d508bac10ef9814bba0a8c9f888ff52cee2de651c0`
Drone Log Name: FLY006.csv
Start: 2017-06-29T20:18:32.862000+00:00
End: 2017-06-29T20:19:12.060000+00:00
Duration: 39.2 s

Flight Metrics

Peak Height: 0.056174768385403695 m
Distance Travelled: 2.089905326409388 m
Video Recording: No
Photo Capture: No

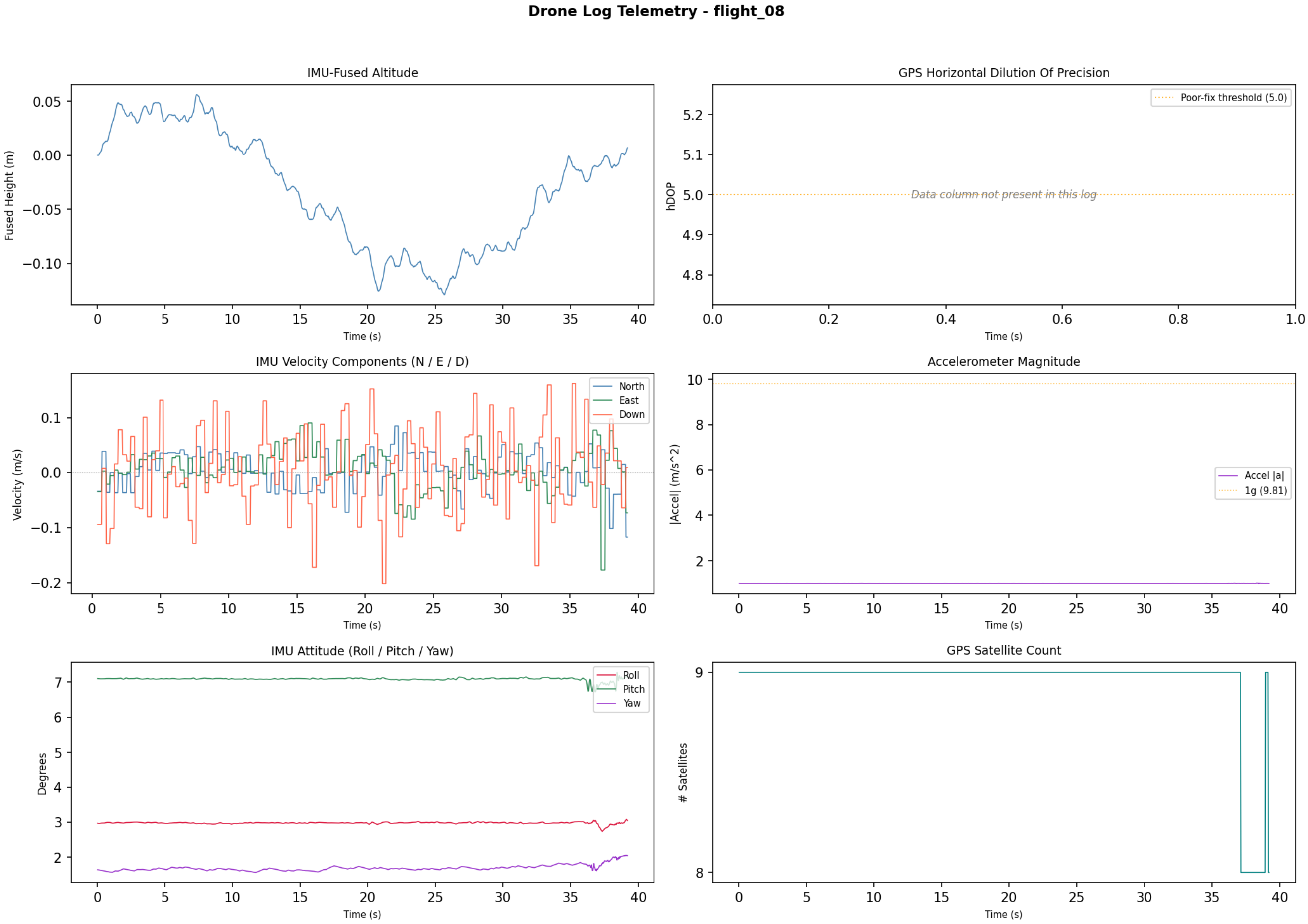
Fly Modes Observed (Chronological)

1. 0

Key Events

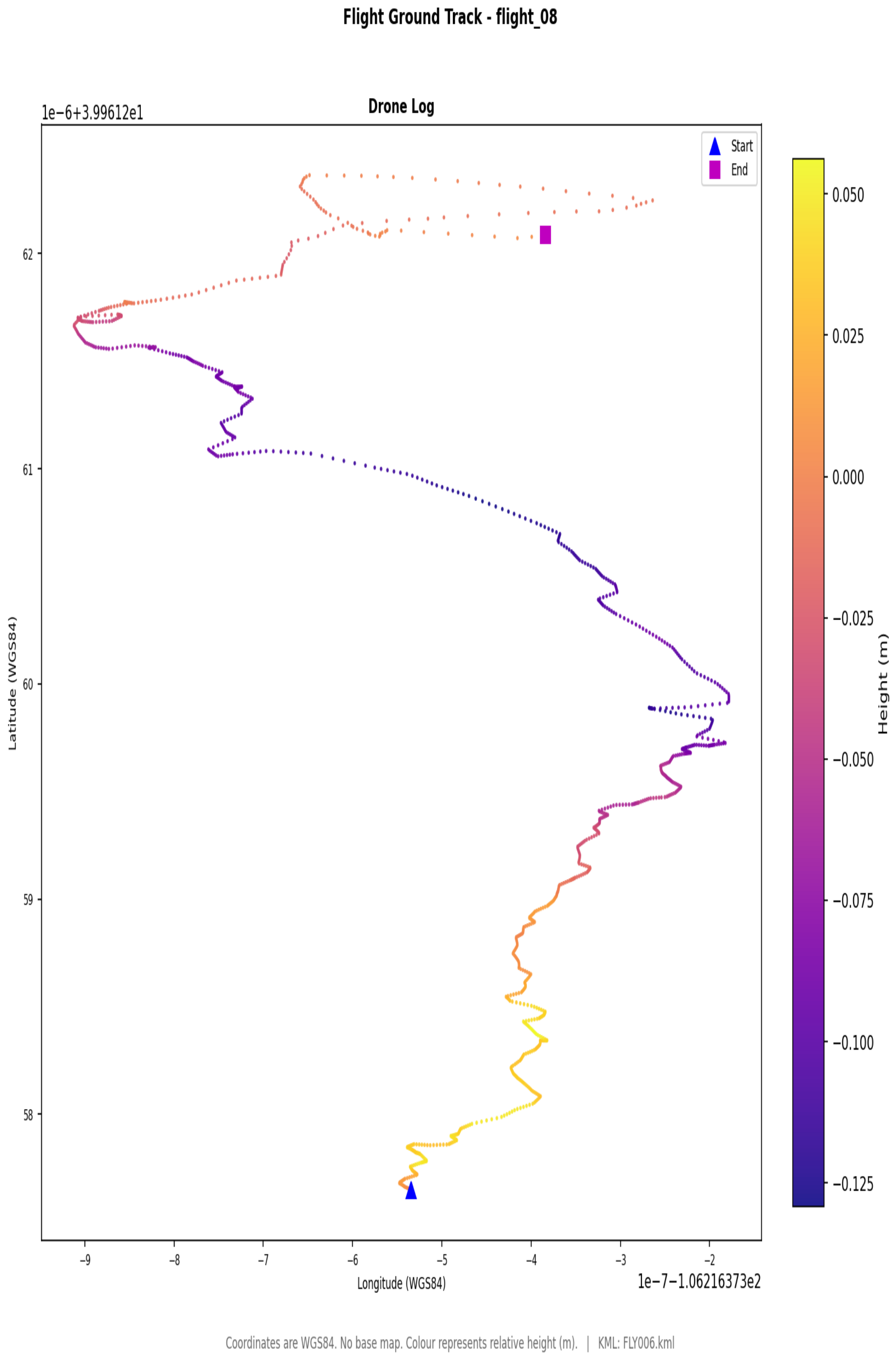
Timestamp (UTC)	Source	Event	Details
2017-06-29 20:18:33	drone_flight_storage:dl	Log started	P3I1
2017-06-29 20:18:40	drone_flight_storage:dl	Reached peak height	0.1 m
2017-06-29 20:19:12	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (metres).
A KML file (FLY006.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

5. Artefact Coverage

Source Coverage

Source	Detected
controller_android	No
controller_ios	Yes
drone_sd	Yes
drone_flight_storage	Yes

Artefact Counts per Source and Category

Source	Account Data	Databases	Drone Logs	Flight Logs	Flight Records	Images	Videos
controller_ios	2	12	0	4	2	1	2
drone_sd	0	0	0	0	0	1	1
drone_flight_storage	0	0	7	0	0	0	0

Data Quality Notes per Category

Databases (12 files):

- 3 database(s) empty (controller_ios)

Drone Logs (7 files):

- 7 file(s) with row anomalies
- 7 file(s) with empty columns
- 7 file(s) with constant-value columns

Flight Logs (4 files):

- 1 file(s) unreadable (binary)
- formats: 2 bracketed_log, 1 json_array_log

Flight Records (2 files):

- 2 file(s) with row anomalies
- 2 file(s) with empty columns
- 2 file(s) with constant-value columns

Images (2 files):

- 2 file(s) missing (normalisable) EXIF timestamp
- 2 file(s) missing EXIF GPS coordinates

Videos (3 files):

- 1 file(s) contain(s) EXIF GPS coordinates
- 2 file(s) missing (normalisable) EXIF timestamp
- 1 file(s) missing EXIF GPS coordinates

6. Data Quality and Anomalies

Data Quality Observations

- Drone_logs: 7 file(s) with row anomalies detected in telemetry data.
- Drone_logs: 7 file(s) with empty columns - some telemetry channels contain no data.
- Drone_logs: 7 file(s) with constant-value columns - some telemetry channels may be unreliable.
- Flight_logs: 1 file(s) unreadable (binary) - file content could not be parsed.
- Flight_logs: log files present in the following format(s): 2 bracketed_log, 1 json_array_log.
- Flight_records: 2 file(s) with row anomalies detected in telemetry data.
- Flight_records: 2 file(s) with empty columns - some telemetry channels contain no data.
- Flight_records: 2 file(s) with constant-value columns - some telemetry channels may be unreliable.

Pipeline Anomaly Flags

1. [p2 - controller android]: source evidence not found
2. [p3 - controller android]: source evidence not found
3. [p3 - controller ios - drone logs]: no artefacts found
4. [p3 - drone flight storage - drone logs]: empty file moved to _rejected: FLY007.DAT
5. [p4 - controller android]: source evidence not found

Artefact-Level Anomaly Detail

The following summarises which automated checks fired for each artefact processed in Phase 5. Row counts indicate affected data rows; column lists are truncated to five entries.

Databases

djiFMDB.db
- database_empty: all tables empty

djiFMDB_1.db
- database_empty: all tables empty

statistics.db
- database_empty: all tables empty

Drone Logs

norm_FLY000.csv
- missing_gps: 19008 row(s) affected
- contains_no_value: 27 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+22 more)
- contains_constant_value: 61 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+56 more)

norm_FLY001.csv
- missing_gps: 17641 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+21 more)
- contains_constant_value: 60 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+55 more)
- quaternion_invalid: 171 row(s) affected

norm_FLY002.csv
- missing_gps: 15552 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+21 more)
- contains_constant_value: 58 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+53 more)
- quaternion_invalid: 170 row(s) affected
- attitude_out_of_bounds: 262 row(s) affected

norm_FLY003.csv

- missing_gps: 11137 row(s) affected
- contains_no_value: 27 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+22 more)
- contains_constant_value: 48 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+43 more)
- quaternion_invalid: 4 row(s) affected

norm_FLY004.csv

- missing_gps: 29362 row(s) affected
- altitude_negative: 94 row(s) affected
- contains_no_value: 10 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+5 more)
- contains_constant_value: 14 column(s) - AirCraftCondition, Battery(0):maxVolts, GPS(0):numGLNAS, GPS(0):numGPS, HP:rthHeight (+9 more)
- quaternion_invalid: 257 row(s) affected

norm_FLY005.csv

- missing_gps: 46089 row(s) affected
- altitude_negative: 19349 row(s) affected
- contains_no_value: 10 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+5 more)
- contains_constant_value: 14 column(s) - AirCraftCondition, Battery(0):maxVolts, GPS(0):numGLNAS, GPS(0):numGPS, HP:rthHeight (+9 more)
- quaternion_invalid: 321 row(s) affected
- attitude_out_of_bounds: 95 row(s) affected

norm_FLY006.csv

- missing_gps: 1121 row(s) affected
- altitude_negative: 148 row(s) affected
- contains_no_value: 11 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+6 more)
- contains_constant_value: 52 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+47 more)

Flight Logs

2017-06-29_10_30_49.dat

- format: json_array_log | entries: 1

upgradeLog_0.txt

- format: bracketed_log | entries: 150

upgradeLog_1.txt

- format: bracketed_log | entries: 27

Flight Records

norm_DJIFlightRecord_2017-06-29__10-30-47_.csv

- altitude_negative: 146 row(s) affected
- contains_no_value: 41 column(s) - APP_GPS.accuracy, APP_GPS.latitude, APP_GPS.longitude, APP_SER_WARN.warn, APP_WARN.warn (+36 more)
- contains_constant_value: 183 column(s) - APP_TIP.tip, CAMERA_INFO.cameraType, CAMERA_INFO.connectState, CAMERA_INFO.enabledPhoto, CAMERA_INFO.encryptStatus (+178 more)

norm_DJIFlightRecord_2017-06-29__10-32-33_.csv

- altitude_negative: 106 row(s) affected
- altitude_spike: 5 row(s) affected
- contains_no_value: 41 column(s) - APP_GPS.accuracy, APP_GPS.latitude, APP_GPS.longitude, APP_SER_WARN.warn, APP_TIP.tip (+36 more)
- contains_constant_value: 174 column(s) - CAMERA_INFO.cameraType, CAMERA_INFO.connectState, CAMERA_INFO.enabledPhoto, CAMERA_INFO.encryptStatus, CAMERA_INFO.firmErrorType (+169 more)

Images

2017_06_29_10_33_22.thumbnail

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

Videos

2017_06_29_10_33_22.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

7. Further Investigation Items

The following items were identified by automated analysis as requiring manual investigator follow-up. They represent unresolved questions or limitations that the pipeline cannot resolve automatically.

1. Drone Model inconsistency detected - conflicting values observed: P3 Standard, P3I1, P3S. Manual verification required.
2. Controller Device inconsistency detected - conflicting values observed: iPad5,1__weibosdk__003143000__iphone__os10.3.1, iPad5,1__weibosdk__003143000__iphone__os10.3.2. Manual verification required.
3. DJI Application Version inconsistency detected - conflicting values observed: 3.1.12, 4.1.3. Manual verification required.
4. Product Type inconsistency detected - conflicting values observed: 2, 7. Manual verification required.
5. Note: no firmware version found - Device and account information.
6. Last App Launch inconsistency detected - conflicting values observed: 2017-06-28T19:40:33Z, 2017-06-28T20:20:50Z. Manual verification required.
7. Note: no last known latitude (find aircraft) found - Device and account information.
8. Note: no last known longitude (find aircraft) found - Device and account information.
9. Note: no last known altitude (find aircraft) found - Device and account information.
10. Note: no last known heading (find aircraft) found - Device and account information.
11. Note: no last known horiz. accuracy (find aircraft) found - Device and account information.
12. 7 flight record(s) could not be correlated to a drone log (flight_01, flight_02, flight_03, flight_04, flight_06, flight_07, flight_08). Independent corroboration is absent for these flight(s).
13. 9 database file(s) contain records and should be examined for relevant artefacts.

Appendix A - Extracted Artefact Hash Manifest

Each artefact extracted in Phase 3 is listed below with its full SHA-256 and SHA-1 cryptographic hashes. These values can be used to independently verify the integrity of every extracted file.

com.dji.go.plist	**Account Data**	**controller_ios**	**extract_logical**
SHA-256:	018095a4f11cccba2e3e35ae631e458878a2644cbf24ef6a6d28077f1d86977f		
SHA-1:	f9eeee2d16e9d28da2cded5b34ccd0803a3e4dd4		

com.dji.pilot.plist	**Account Data**	**controller_ios**	**extract_logical**
SHA-256:	07ef7d50194dcbc01a985e6c1955c3286bfa35eb86ab107c220848b2623881cc		
SHA-1:	dbe7177a221d421f7492538b737276f714fa4662		

airmap_flysafeplaces.db	**Databases**	**controller_ios**	**extract_logical**
SHA-256:	d02f441ba79d6c82b33db8c2eca742ffc278bb998eb46edaa57290f53b992ee6		
SHA-1:	805373eb7dd57a932cc3438234763f1a27b61a9c		

airmap_flysafeplaces_1.db	**Databases**	**controller_ios**	**extract_logical**
SHA-256:	01472f913caf3ce5d50b146cd37a6fea1a18ca4d9666b2d5dc06189a8ac5c17c		
SHA-1:	fdecfca00525fd0f36dc51b5a8709efaf409f908		

djiFMDB.db	**Databases**	**controller_ios**	**extract_logical**
SHA-256:	baa7277998f1c5db44be2b44a2264bedd94f93495fbb0d73636ee2fdc9c94848		
SHA-1:	d6720f20112c173533a7ad8ccedd8c8992503ddb		

djiFMDB_1.db	**Databases**	**controller_ios**	**extract_logical**
SHA-256:	003687bc29cf9f101fae000e55fef1718de111f65242f517487ab2cb826a3e37		
SHA-1:	86fe40031cfeab9bdb5627a6b0bb7a446abb05f1		

nfz_flysafeplaces.db	**Databases**	**controller_ios**	**extract_logical**
SHA-256:	74e9055812a04a2b5e561094b52c26203180aba76f44d398558559b537f22248		
SHA-1:	e04a79802f2dfc3d18c519dfa532cfe6995d54d4		

nfz_flysafeplaces_1.db	**Databases**	**controller_ios**	**extract_logical**
SHA-256:	dc183c19fddcf46e45ebee249f98366b379e2deb16723931865952be7e07de94		
SHA-1:	eb80a01b91afc01f0d9dab26112c078fced6fc1d		

statistics.db	**Databases**	**controller_ios**	**extract_logical**
SHA-256:	83bfc672a35016dcdfa168cbc780d6219345ddfe601ad75087ce879503a3058a		
SHA-1:	3a8843449db0f78d4502b81d67e0f26584226a80		

statistics_1.db	**Databases**	**controller_ios**	**extract_logical**
SHA-256:	1a6de1261c7894f8c5838045f701f51171a4b4b4175bc941c6aa1809d412c794		
SHA-1:	b41f924e4c9a0b34dab68cea3eff5e0dd129793f		

tencent_analysis.db	**Databases**	**controller_ios**	**extract_logical**
SHA-256:	91a05a0acc06a3d050cda7dd84de0d69273a5fee2c895d5ec9acc956801fe81d		
SHA-1:	94c90db3c6f4a2815a3538e4abae3a8fe7329bd4		

tencent_analysis_1.db	**Databases**	**controller_ios**	**extract_logical**
SHA-256:	377a39781bcad779230cd58ca0dd1bfa8aad9c082c2dcd7dc49e158d8f4f2b41		
SHA-1:	714fda8433de28bec957523e505414453d562540		

tencent_wxo_analysis.db		Databases	controller_ios	extract_logical
SHA-256:	d8c4dffacf1a1fc3a3aa5ef4d240f2a76598a337356b08c0aa84e936e5e93b9f			
SHA-1:	0d33194668dc7733e179559c0cf9d457d93e3adf			

tencent_wxo_analysis_1.db		Databases	controller_ios	extract_logical
SHA-256:	212db40de8c90f74bde4109c96ff47798a75f91cafeab1f34ebab5f15f19dd40			
SHA-1:	f3ccf41369475c57c8b658e36f6c48fc05668e66			

FLY000.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	f967f84868805990c5a4d48239c8d50577d6bc026cdf6d8298172e917506cd94			
SHA-1:	fe0594c794a3e85fc74614bdb61a5ae6bd699656			

FLY001.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	8676d51b5aad573bca86c5f12d714075a355408579afa70d6e9d5ca7af759bab			
SHA-1:	64aaa3aacb4faabdf4c71b7ec641b87dc3fbf44b			

FLY002.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	3afd2bd6f2a135ced16665602085dcc2df4c2f36f118d91db9e1f310c5b1d585			
SHA-1:	e0a7e2c61e13ff4e715e3e3b241f37c2797ecae1			

FLY003.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	a158c39b5d139fe929587c17d7e76cdf5b93a0bde5ec9e895aee04b461dcee56			
SHA-1:	10e4301ac71fc9d2986169067a941cd2bddf4b69			

FLY004.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	856a3070d25eca1c431bed4141b78db1074f521247f474000db331d727c67710			
SHA-1:	fd79869ceba8eaecf2fc6ea6a8d6e3b819bde49d			

FLY005.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	692af27bc9c783817d2356e45269ecd9618362e549acf546b8f3bdbea52b4359			
SHA-1:	e7383bfb543145cc4978f969180a9c203aa553e3			

FLY006.DAT		Drone Logs	drone_flight_storage	extract_physical
SHA-256:	20005e5c73a3c2e7b20ee3a1d8354d4c81b558c40cc84a60d677a85382fdc955			
SHA-1:	155379175baa7dfea991c52c94eafbdec9dd02a1			

2017-06-29_10_30_49.dat		Flight Logs	controller_ios	extract_logical
SHA-256:	6fd259eca6ae9d04179d03d279dbf8e30e5e5ae90d52ed06e0c13a8f453a29a2			
SHA-1:	452d0c0f56b9482d75ae8b9517103b3b92085761			

app_log.txt		Flight Logs	controller_ios	extract_logical
SHA-256:	00cc1065cb3d1dced6e3246fbf73c1294524a0bd8340ff2aee579a12efd16498			
SHA-1:	42aa12fbe4cceb7520e9218f488445d42cd1bbf1			

upgradeLog_0.txt		Flight Logs	controller_ios	extract_logical
SHA-256:	2f4b1ea4be58401888eac01c4a9350246675bfb23795a321f1f1b9b1956788c8			
SHA-1:	4f63c2d51d328cc75841cf6963cd39ee243e6b8a			

upgradeLog_1.txt		Flight Logs	controller_ios	extract_logical
SHA-256:	bacadedf528eaffabdd08abce38b8aa09fd36141381daa56b64084a30ae74f1c			
SHA-1:	97d6a9930c220d1eb77e5c0a5a97a6cc74579102			

DJIFlightRecord_2017-06-29__10-30-47_.txt	Flight Records	controller_ios	extract_logical
SHA-256:	e20134ca41c3aad7b88485d606b35e87c78b527710afff29a628716a3f5a4ae4		
SHA-1:	667e643291e2d8dfd574aa26c2f318fea4c8d09c		

DJIFlightRecord_2017-06-29__10-32-33_.txt	Flight Records	controller_ios	extract_logical
SHA-256:	1aee75e5afd15a043980433e777e0a731d64525d9952f079aeee71def19d8c36		
SHA-1:	86de50dd6540c803b27d21198871ef8dcfd88702		

2017_06_29_10_33_22.thumbnail	Images	controller_ios	extract_logical
SHA-256:	9e3318a313c4181985a2d1117cae0838fc9cd172720272a98ec505c448056803		
SHA-1:	b4f97364b057aa5af8d1148200f8d818171db21c		

DJI_0001.THM	Images	drone_sd	extract_physical
SHA-256:	225ae395fd38565453e36f6ef195a650fffb9d04b32e1f5bc2fb66960532ca32		
SHA-1:	f5a21ab7775435b4d78b22c067a8a638db0509f5		

2017_06_29_10_33_22.info	Videos	controller_ios	extract_logical
SHA-256:	848af19a75fe78c2aacb7f286847a75a957da52b73d523be88ecc45ce25ecb71		
SHA-1:	0aaf14dd59d14c367809854c1727e61d495e47fa		

2017_06_29_10_33_22.mp4	Videos	controller_ios	extract_logical
SHA-256:	738ee5be9a54f3d0ec4a6159acdc9435ef017f2be5698b6f8528b7a7b924e50f		
SHA-1:	ed3108801da503948cc53818a7e8be943846ebd1		

DJI_0001.MOV	Videos	drone_sd	extract_physical
SHA-256:	e034453ea8d0e3db7eaf530cced43bcbc7036483b9e01ccb19fbfd2d6287c6c7		
SHA-1:	54fd3ce809c92854bc5c6d95a641893828fc3b2c		

Appendix B - Evidence Derivation Summary

This table summarises how many artefacts were derived at each pipeline phase, per evidence source and category. It demonstrates the processing chain from extracted artefacts (P3) through decoded outputs (P4) to normalised analysis-ready files (P5).

Controller Ios

Category	P3 Extracted	P4 Decoded	P5 Normalised
Account Data	2	2	0
Databases	12	0	0
Flight Logs	4	0	0
Flight Records	2	2	2
Images	1	1	1
Videos	2	2	0

Drone Sd

Category	P3 Extracted	P4 Decoded	P5 Normalised
Images	1	1	0
Videos	1	1	0

Drone Flight Storage

Category	P3 Extracted	P4 Decoded	P5 Normalised
Drone Logs	7	14	7

Appendix C - Tool Invocation Log

All external tool invocations executed during this analysis are recorded below, in chronological order. Each entry includes the timestamp, tool name, version, return code, full command, and output files produced.

2026-06-18 10:12:52 mmls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: mmls.exe df003_external.001

2026-06-18 10:12:57 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p -o 8192 df003_external.001

2026-06-18 10:12:57 mmls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: mmls.exe df003_internal.001

2026-06-18 10:13:00 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p -o 63 df003_internal.001

2026-06-18 10:13:43 parser_ios v- rc=n/a

cmd: df003_flight.zip C:\Users\Floris\Documents\dfdof_output\ft-df003-2017-ios\p2_image_parsing\controller_ios_parsed

out: controller_ios_parsed

2026-06-18 10:13:53 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df003_external.001 40965

out: DJI_0001.MOV

2026-06-18 10:14:01 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df003_external.001 602117

out: DJI_0001.THM

2026-06-18 10:14:01 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df003_internal.001 5

out: FLY000.DAT

2026-06-18 10:14:02 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df003_internal.001 6

out: FLY001.DAT

2026-06-18 10:14:02 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df003_internal.001 7

out: FLY002.DAT

2026-06-18 10:14:03 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df003_internal.001 8

out: FLY003.DAT

2026-06-18 10:14:04 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df003_internal.001 9

out: FLY004.DAT

2026-06-18 10:14:05 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df003_internal.001 10

out: FLY005.DAT

2026-06-18 10:14:06 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df003_internal.001 11

out: FLY006.DAT

2026-06-18 10:14:06 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df003_internal.001 12

out: FLY007.DAT

2026-06-18 10:14:07 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTlogToCSVtool.exe DJIFlightRecord_2017-06-29__10-30-47_.txt DJIFlightRecord_2017-06-29__10-30-47_.csv *out: DJIFlightRecord_2017-06-29__10-30-47_.csv*

2026-06-18 10:14:08 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTlogToCSVtool.exe DJIFlightRecord_2017-06-29__10-32-33_.txt DJIFlightRecord_2017-06-29__10-32-33_.csv

out: DJIFlightRecord_2017-06-29__10-32-33_.csv

2026-06-18 10:14:08 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_10_33_22.thumbnail

out: 2017_06_29_10_33_22_exif.json

2026-06-18 10:14:09 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_10_33_22.mp4

out: 2017_06_29_10_33_22_exif.json

2026-06-18 10:14:09 exiftool v13.57 rc=1

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.THM

out: DJI_0001_exif.json

2026-06-18 10:14:10 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.MOV

out: DJI_0001_exif.json

2026-06-18 10:14:48 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY000.DAT

out: FLY000.csv, FLY000.kml

2026-06-18 10:15:08 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY001.DAT

out: FLY001.csv, FLY001.kml

2026-06-18 10:15:24 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY002.DAT

out: FLY002.csv, FLY002.kml

2026-06-18 10:15:48 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY003.DAT

out: FLY003.csv, FLY003.kml

2026-06-18 10:16:05 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY004.DAT

out: FLY004.csv, FLY004.kml

2026-06-18 10:16:25 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY005.DAT

out: FLY005.csv, FLY005.kml

2026-06-18 10:16:40 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY006.DAT

out: FLY006.csv, FLY006.kml

Appendix D - DFDOF Proof of Concept Overview

The Drone Forensic Decision and Orchestration Framework (DFDOF) is an automated 8-phase forensic pipeline for DJI drone evidence. It ingests images of the controller, drone flight storage and SD cards, then processes them into a structured analysis and report.

P1 Provenance and Integrity

Classifies input files (ZIP/E01/.001) by content signatures. Computes SHA-256 and SHA-1 for all input evidence. Presents a source classification summary for operator confirmation before processing begins.

Input: evidence directory. Output: classified evidence objects with hashes.

P2 Image Parsing

Makes all sources search-ready. Decrypts iOS backups into a domain directory tree. Extracts Android device metadata (DeviceInfo.xml, ApplicationInfo.xml, packages.list). Produces backup_info.json with device identifiers and installed DJI applications.

Input: raw evidence files. Output: parsed directory structures, backup_info.json.

P3 Artefact Extraction

Root-driven category extraction. Discovers DJI app roots by bundle-ID segment matching. Recurses into configured per-platform category paths and filters by extension. Rejects empty files. Each artefact is wrapped in an Evidence object with a parent hash linking it to its source.

Input: parsed directories. Output: categorised artefacts (flight records, drone logs, flight logs, databases, images, videos, account data).

P4 Decision and Orchestration

Dispatches each artefact category to the appropriate specialist tool: DatCon for .DAT drone logs (produces CSV + KML), TXTlogToCSV for flight record .txt files, ExifTool for images and videos, Python parsers for account data (.plist/.xml). Computes hashes for all tool outputs.

Input: extracted artefacts. Output: decoded CSVs, KML files, EXIF JSON, account data observations.

P5 Normalisation and Anomaly Checking

Normalises all CSVs to UTC timestamps and a unified schema. Applies 9 shared telemetry anomaly checks (timestamp regression/gaps, zero coordinates, GPS quality, altitude spikes, motor-airborne-off) plus format-specific checks for flight records (battery voltage, temperature) and drone logs (quaternion, hDOP). Detects empty databases, unreadable flight logs, and missing (normalisable) EXIF timestamps.

Input: decoded artefacts. Output: normalised CSVs, anomaly observations.

P6 Multi-Source Correlation

Correlates flight records with drone logs using two criteria: (1) temporal overlap >= 60 s AND >= 90% of the shorter flight's duration; (2) median haversine distance of nearest-neighbour GPS pairs <= 25 m with >= 5 matched pairs. Deduplicates FR candidates when multiple apps recorded the same flight. Builds a unified per-flight event timeline from all matched sources. Also correlates media files by EXIF timestamp, GPS bounding box, and video duration. In the correlation summary: Overlap is the temporal overlap as a percentage of the shorter flight's duration; GPS Match Rate is the fraction of flight-record GPS rows that found a spatially matched drone-log row within a +-2 s window.

Input: normalised CSVs. Output: timeline_flightXX.json per flight.

P7 Analysis and Validation

Aggregates results from all prior phases into a structured analysis. Resolves device identity (drone serial, model, app version, account email) across sources with confidence levels (controller-only / corroborated / inconsistent). Computes per-flight metrics (peak height, media capture, record completeness). Produces coverage scores and a grouped conclusions structure with a dedicated further_investigation section.

Input: all phase outputs. Output: analysis dict with conclusions in state.json.

P8 Automated Reporting

Generates this PDF forensic baseline report from all prior phase outputs. Produces sensor telemetry plots (altitude, speed, attitude, battery, GPS) from normalised CSVs, a dual-source flight track plot, and the structured appendices. All figures are also saved as standalone PNG files.

Input: state.json (all phase outputs). Output: PDF report + PNG plot files.

Disclaimer

DFDOF was developed by Floris Krijger, MSc Security and Network Engineering, University of Amsterdam, as a Master's thesis project (1 April 2026 - 30 June 2026). This proof of concept has been developed and tested on a targeted subset of the NIST VTO forensic drone dataset and is not a finished, production-ready system. Before use in any operational or legal context, further testing, validation across additional drone models and evidence types, and robustness improvements are required. The authors accept no liability for conclusions drawn from automated pipeline output without independent expert verification.